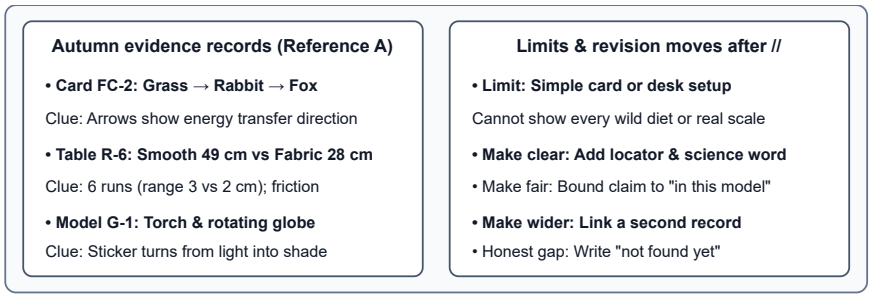
Reference sheets

Reference A · Autumn science evidence records, clues, and limits

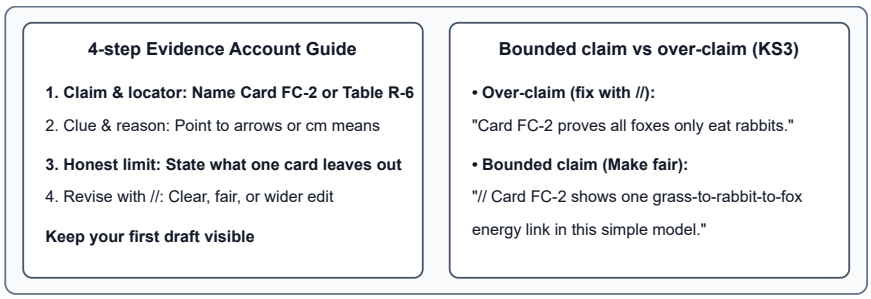


Autumn evidence records table. It shows Card FC-2, Table R-6, Model G-1, limits, and revision moves.

Reference only. Do not write on this sheet. Use these locators and clues to support your science account.

Locator code.	Observed clue on record.	Key Stage 3 science meaning.	Honest model or test limit.
1. Card FC-2 (Food-chain diagram).	Labelled arrows point from Grass to Rabbit and from Rabbit to Fox.	Arrows show energy transfer from producer to primary and secondary consumers.	Shows one three-species chain; omits energy amounts and other wild prey.
2. Table R-6 (Rover six-run table).	Smooth: 48, 51, 49 cm (mean 49, range 3). Fabric: 27, 29, 28 cm (mean 28, range 2).	Lower contact friction on smooth surface opposed rover motion less than fabric.	Tested only one unpowered classroom toy across two flat indoor strips.
3. Model G-1 (Torch and turning globe).	A marked spot on the rotating ball turns from the torch beam into shadow.	Earth's rotation on its axis brings the daily cycle of day and night.	The torch and classroom ball are not to real astronomical size or distance.

Reference B · Four-step Evidence Account frame and // revision moves



Four-step Evidence Account guide. It shows linking locators, stating limits, and making // revisions.

Revision move.	Sentence stem for your book.	What to check before and after //.
1. Make clear (Add locator & term).	On [locator], the clue ___ shows ___.	Did you name Card FC-2, Table R-6, or Model G-1 and the science word?
2. Make fair (Bound the claim).	// In this model (or in this test), ___ shows ___, but cannot show ___.	Did you narrow 'all/everywhere' to what this record actually shows?
3. Make wider (Link a 2nd record).	// A second autumn record, [locator], also shows ___ because ___.	Did you connect a second real record from Reference A?
4. Honest gap (No guessing).	If a source or measurement is missing, write 'not found yet'.	Did you stay honest instead of inventing a number?

Made by Matt

Tier pack · Aspire

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use Box 1 of the arrival drawing. Explain the energy role of grass, rabbit, and fox on Card FC-2.

Help. Link producer and consumers to the energy arrows.

2. Compare a bounded claim with an over-claim when using classroom science records.

Help. Explain why bounding claims to the evidence matters.

3. Look at Box 2 and Reference A. Contrast the evidence in Table R-6 (49 cm vs 28 cm) with Card FC-2.

Help. Contrast measured cm numbers with a directional model diagram.

4. What should a scientist record if a second source is missing during a recap?

Help. Honest reporting protects scientific trustworthiness.

You do tasks

1. Use Reference A. Write a comparative account linking two autumn records. Include exact locators, clues, and mechanisms. **Staff I ☐ S ☐ C ☐**

Help. Name each locator, its clue, and the mechanism (such as energy transfer or friction).

2. Evaluate the limits of both chosen records. Explain why bounding claims to the evidence prevents scientific over-claiming. **Staff I ☐ S ☐ C ☐**

Help. Contrast what each record shows with what it omits.

3. Use Reference B. Complete a two-part // revision. Narrow an over-claim (Make fair) and link a second locator (Make wider). **Staff I ☐ S ☐ C ☐**

Help. Keep the draft visible and add both revision moves after //.

Optional participation-loop prompt. Share your evidence claim and limit with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Why does every scientific evidence account need both a locator and a bounded claim?

2. Contrast the model limit of Card FC-2 with the limit of Model G-1.

3. Why is writing 'not found yet' better science than guessing a missing measurement?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Aspire

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt